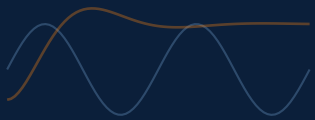
DAY 2 / 5

Mechatronics: Electrical Machines, Control, Servo Systems and Sensing



# System Dynamics and Control

**Linearisation ▪ Transfer functions & state space ▪ Identification ▪ PID ▪ Stability**

Mechatronics Training Programme

Theoretical Course Series ▪ Day 2 of 5

# Agenda and learning objectives

| Code | Session | Time |
|---|---|---|
| 2.1 | Nonlinear effects and linearisation (cont.) | 0.5 h |
| 2.2 | Transfer functions and state space | 1.0 h |
| 2.3 | System identification and frequency response | 1.0 h |
| 2.4 | PID and basic control structures | 1.5 h |
| 2.5 | Stability and performance analysis | 1.0 h |
| | **Net teaching time** (+ short breaks) | **5.0 h** |

**After today you can . . .**

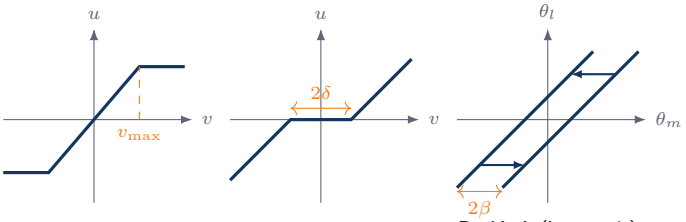
- linearise a nonlinear servo model about an operating point
- move freely between $G(s)$, $(A, B, C, D)$ and block diagrams
- identify FOPDT / 2nd-order models from steps and FRFs
- tune and implement a practical PID with anti-windup, feedforward and cascades
- judge stability and robustness via PM, GM, $M_s$, $S$ and $T$

**Notation for the whole day:** plant $G(s)$, controller $C(s)$, loop transfer function $L(s) = C(s)G(s)$, sensitivity $S = 1/(1 + L)$, complementary sensitivity $T = L/(1 + L)$, crossover $\omega_c$, phase margin PM, gain margin GM. Motor: $R, L_a, k_e, k_t, J, B, T_L$ (armature inductance written $L_a$ to avoid a clash with $L(s)$).

Session 1

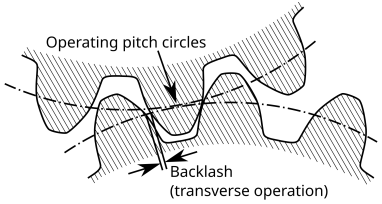
# 2.1 Nonlinear Effects and Linearisation

# Static nonlinearities in real servo axes




**Saturation** **Dead-zone** **Backlash (hysteresis)**

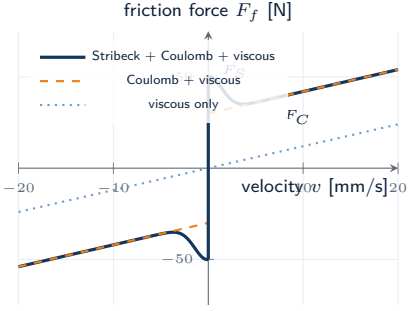
| Nonlinearity | Where in a servo | Typical symptom |
|---|---|---|
| Saturation | DC-link voltage, current limit | windup, slow large steps |
| Dead-zone | IGBT dead-time, valve overlap | distortion near zero, limit cycles |
| Backlash | gears, couplings, belts | position error at reversal, chatter |




GearHeads, Slashme, CC BY-SA 3.0, via Wikimedia Commons

Saturation: $u = \mathrm{sat}(v) = \max(-v_{\max}, \min(v, v_{\max}))$.
Static map $\Rightarrow$ the **describing function** $N(A)$ (amplitude-dependent gain) is the classical tool to predict limit cycles [Slotine & Li, 1991].

# Friction: Coulomb, viscous and Stribeck

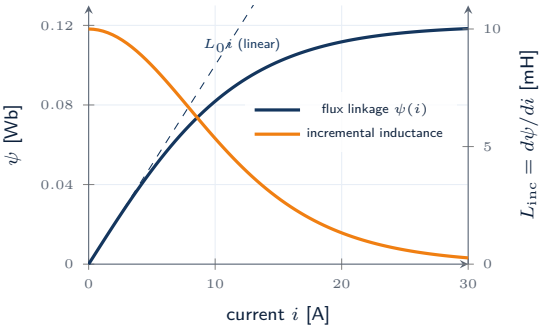



$$F_f(v) = \left[F_C + (F_S - F_C)\, e^{-(v/v_S)^2}\right] \operatorname{sgn}(v) + B\, v$$

- $F_S$ static (breakaway), $F_C$ Coulomb level, $v_S$ Stribeck velocity, $B$ viscous coefficient.
- At $v = 0$ the force is **set-valued** in $[-F_S, F_S]$ (stiction) $\Rightarrow$ non-smooth, not linearisable.
- Negative slope $dF_f/dv < 0$ at low speed acts as **negative damping** $\Rightarrow$ stick–slip.
- Integral action + stiction $\Rightarrow$ **hunting limit cycles** around the set-point.
- Velocity reversal $\Rightarrow$ **quadrant glitches** in contouring (CNC circles).
- Remedies: friction feedforward $\hat{F}_C \operatorname{sgn}(v_{\mathrm{ref}})$, dither, LuGre observer.

[Armstrong-Hélouvry, Dupont & Canudas de Wit, Automatica 1994]

# Magnetic saturation: parameters that depend on the operating point




Iron saturates: $\psi(i)$ bends over and the **incremental inductance** falls,

$$v = R\,i + \frac{d\psi}{dt} = R\,i + L_{\mathrm{inc}}(i)\,\frac{di}{dt}, \qquad L_{\mathrm{inc}} = \frac{\partial \psi}{\partial i}.$$

- Current-loop plant $1/(L_{\mathrm{inc}}s + R)$: gain **rises** as $L_{\mathrm{inc}}$ drops $\Rightarrow$ crossover moves up, phase margin shrinks at overload.
- Torque constant $k_t$ falls at peak current (PMSM: $\psi_{\mathrm{PM}}$ + armature reaction).
- Example curve: $\psi = 0.12\tanh(i/12)$ Wb $\Rightarrow$ $L_{\mathrm{inc}}$: 10 mH at 0 A, 2.8 mH at 24 A.
- Practice: gain scheduling $K_p(i) = \hat{L}_{\mathrm{inc}}(i)\,\omega_{cc}$, or design for the worst case.

# Taylor linearisation about an operating point

Nonlinear model $\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{u})$, $\boldsymbol{y} = \boldsymbol{h}(\boldsymbol{x}, \boldsymbol{u})$.
Equilibrium (operating point): $\boldsymbol{f}(\boldsymbol{x}_0, \boldsymbol{u}_0) = \boldsymbol{0}$, $\boldsymbol{y}_0 = \boldsymbol{h}(\boldsymbol{x}_0, \boldsymbol{u}_0)$.
Deviations $\delta\boldsymbol{x} = \boldsymbol{x} - \boldsymbol{x}_0$, $\delta\boldsymbol{u} = \boldsymbol{u} - \boldsymbol{u}_0$; first-order Taylor expansion:

$$\delta\dot{\boldsymbol{x}} \approx A\,\delta\boldsymbol{x} + B\,\delta\boldsymbol{u}, \qquad \delta\boldsymbol{y} \approx C\,\delta\boldsymbol{x} + D\,\delta\boldsymbol{u}$$

$$A = \frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\Big|_0,\ B = \frac{\partial \boldsymbol{f}}{\partial \boldsymbol{u}}\Big|_0,\ C = \frac{\partial \boldsymbol{h}}{\partial \boldsymbol{x}}\Big|_0,\ D = \frac{\partial \boldsymbol{h}}{\partial \boldsymbol{u}}\Big|_0$$

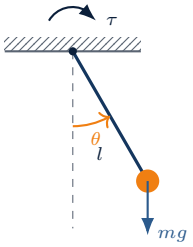
Jacobian entries: $A_{ij} = \partial f_i / \partial x_j$ evaluated at $(\boldsymbol{x}_0, \boldsymbol{u}_0)$.

## Validity – Lyapunov's indirect method

- All eigenvalues of $A$ in the open LHP $\Rightarrow$ equilibrium **locally** asymptotically stable.
- Any eigenvalue in the RHP $\Rightarrow$ unstable.
- Eigenvalue on the $j\omega$-axis $\Rightarrow$ no conclusion from the linear model.

Not applicable at non-smooth points (Coulomb friction at $v = 0$, backlash, saturation corner): use describing functions or simulation. A family of linear models over operating points $\Rightarrow$ **gain scheduling**.

# Example 1: pendulum / unbalanced arm on a servo motor




$$J\ddot{\theta} = \tau - mgl\sin\theta - b\dot{\theta}, \quad \boldsymbol{x} = [\theta,\ \dot{\theta}]^\top,\ u = \tau.$$

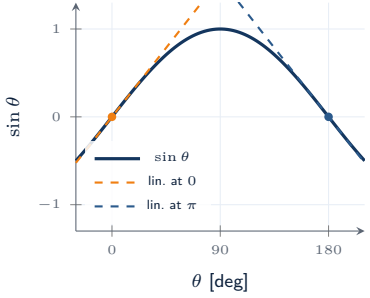
$$A = \begin{bmatrix} 0 & 1 \\ -\frac{mgl}{J}\cos\theta_0 & -\frac{b}{J} \end{bmatrix},\ B = \begin{bmatrix} 0 \\ \frac{1}{J} \end{bmatrix}$$

**Hanging** $\theta_0 = 0$: $\lambda^2 + \frac{b}{J}\lambda + \frac{mgl}{J} = 0 \Rightarrow$ stable, $\omega_n = \sqrt{mgl/J}$.

**Inverted** $\theta_0 = \pi$: $\lambda^2 + \frac{b}{J}\lambda - \frac{mgl}{J} = 0 \Rightarrow$ one **RHP pole**.

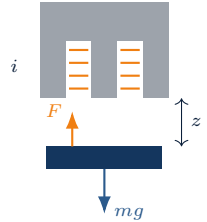
Numbers: $m = 2\,\text{kg}$, $l = 0.3\,\text{m}$, $J = ml^2 = 0.18\,\text{kg m}^2$

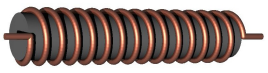
$\Rightarrow \omega_n = \sqrt{g/l} = 5.72\,\text{rad/s}$ (0.91 Hz). Torque needed to hold $\theta_0$: $\tau_0 = mgl\sin\theta_0$ (max 5.9 Nm).




Error of $\sin\theta \approx \theta$: 1 % at 14°, 5 % at 31°.

# Example 2: electromagnet (magnetic levitation / actuator)



Svjo, CC BY-SA 3.0, via Wikimedia Commons

Force (ideal magnetic circuit, gap $z$ measured downward): $F = k\,i^2/z^2$, $m\ddot{z} = mg - k\,i^2/z^2$.

Equilibrium: $i_0 = z_0\sqrt{mg/k}$. Linearise $F(i, z)$:

$$\delta F = \underbrace{\frac{2ki_0}{z_0^2}}_{K_i}\,\delta i - \underbrace{\frac{2ki_0^2}{z_0^3}}_{K_z}\,\delta z \;\Rightarrow\; m\,\delta\ddot{z} = K_z\,\delta z - K_i\,\delta i, \qquad \frac{\delta Z(s)}{\delta I(s)} = \frac{-K_i}{ms^2 - K_z}$$

**Worked example:** $m = 0.1\,\text{kg}$, $z_0 = 5\,\text{mm}$, $i_0 = 1\,\text{A}$
$\Rightarrow k = mgz_0^2/i_0^2 = 2.45 \times 10^{-5}\,\text{N m}^2/\text{A}^2$.

- $K_z = 2mg/z_0 = 392\,\text{N/m}$ (**negative stiffness**), $K_i = 2mg/i_0 = 1.96\,\text{N/A}$.
- Poles $s = \pm\sqrt{K_z/m} = \pm 62.6\,\text{rad/s}$ $\Rightarrow$ open-loop unstable, time constant 16 ms.
- Both gains scale with the operating point: $K_z \propto 1/z_0$, $K_i \propto 1/i_0$ $\Rightarrow$ the linear model is valid only near $(z_0, i_0)$.

Same structure appears in magnetic bearings and in reluctance actuators.

# Session 2.1 – Summary

**Key take-aways**

- Servo axes contain saturation, dead-zone, backlash, friction and magnetic saturation; they set the limits of what linear control can do.
- Friction has a **set-valued** stiction part and a negative-damping Stribeck part: cause of stick–slip and hunting.
- Smooth nonlinearities are handled by **Jacobian linearisation** around an equilibrium $(\boldsymbol{x}_0, \boldsymbol{u}_0)$.
- The linear model is **local**: gains and even stability (pendulum up/down, maglev) depend on the operating point.
- Non-smooth effects need describing functions, compensation (feedforward) or simulation.

Next: having a linear model, we describe it compactly by a transfer function or a state-space model.

Session 2

# 2.2 Transfer Functions and State Space

# Laplace transform, transfer function, poles and zeros

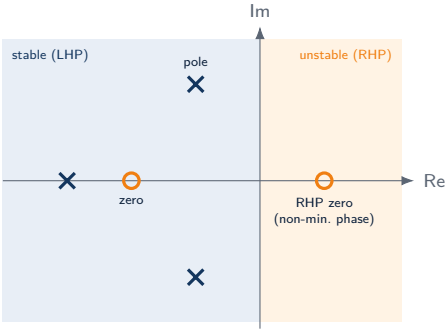
$F(s) = \int_0^\infty f(t)e^{-st}dt$ turns linear ODEs into algebra:

| Time domain | Laplace domain |
|---|---|
| $\dot{f}(t)$ | $sF(s) - f(0)$ |
| $\int_0^t f\, d\tau$ | $F(s)/s$ |
| $f(t - T_d)$ (delay) | $e^{-sT_d}F(s)$ |
| convolution $g * u$ | $G(s)U(s)$ |
| $f(\infty)$ (if it exists) | $\lim_{s\to 0} sF(s)$ |

With zero initial conditions:

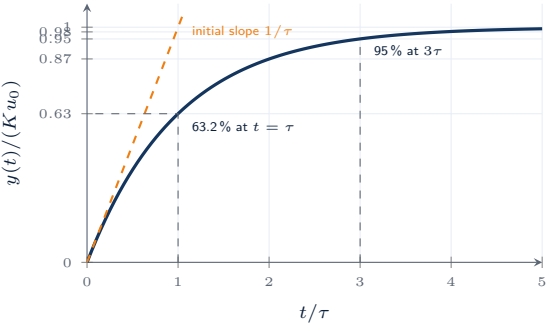
$$G(s) = \frac{Y(s)}{U(s)} = \frac{b_m s^m + \cdots + b_0}{s^n + a_{n-1}s^{n-1} + \cdots + a_0} = K\frac{\prod_i (s - z_i)}{\prod_j (s - p_j)}$$

- **Order** $n$ = number of poles = number of energy stores.
- **Relative degree** $n - m$: proper if $m \le n$ (physical plants: $m < n$).




- **Poles** (roots of denominator) set the *modes* $e^{p_j t}$: stability and speed.
- **Zeros** set how the modes are *weighted*; RHP zeros cause undershoot and limit bandwidth.
- DC gain $G(0)$; time delay $e^{-sT_d}$ is not rational (Padé approximation).

# First-order system




$$G(s) = \frac{K}{\tau s + 1}, \quad y(t) = Ku_0\left(1 - e^{-t/\tau}\right)$$

- One pole at $s = -1/\tau$; bandwidth $\omega_b = 1/\tau$ ($-3$ dB).
- 10–90 % rise time $t_r = 2.2\tau$; 2 % settling $t_s \approx 4\tau$.
- No overshoot, no oscillation – ever.

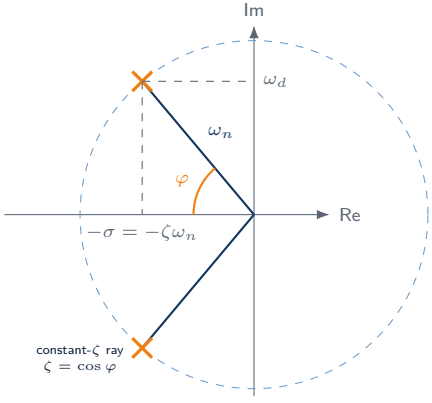
**Servo examples**

- R–L winding: $I/V = 1/(L_a s + R)$, $\tau_e = L_a/R$.
- Inertia with viscous friction: $\Omega/T = 1/(Js + B)$, $\tau = J/B$.
- A well-tuned closed current loop $\approx 1/(\tau_{cl} s + 1)$ seen from the speed loop.

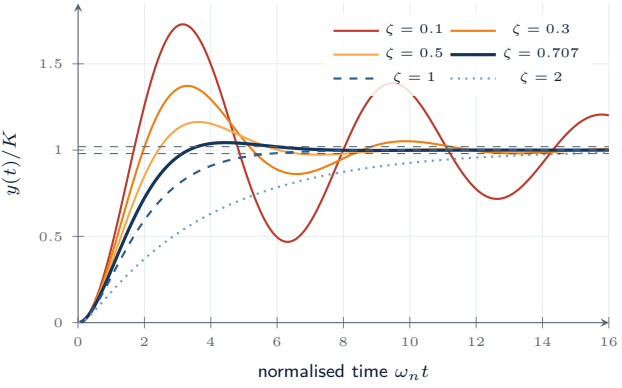
# Second-order standard form and pole geometry

$$G(s) = \frac{K\,\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

- $\omega_n$ natural frequency [rad/s], $\zeta$ damping ratio [–].
- Poles $p_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2} = -\sigma \pm j\omega_d$.
- $|p| = \omega_n$ (distance from origin), $\cos\varphi = \zeta$ (angle to the negative real axis).
- $\zeta < 1$ underdamped, $\zeta = 1$ critically damped, $\zeta > 1$ overdamped (two real poles).
- Mechanical analogue: $m\ddot{x} + c\dot{x} + kx = F \Rightarrow$ $\omega_n = \sqrt{k/m}$, $\zeta = c/(2\sqrt{km})$.



# Step responses of the second-order system for different $\zeta$

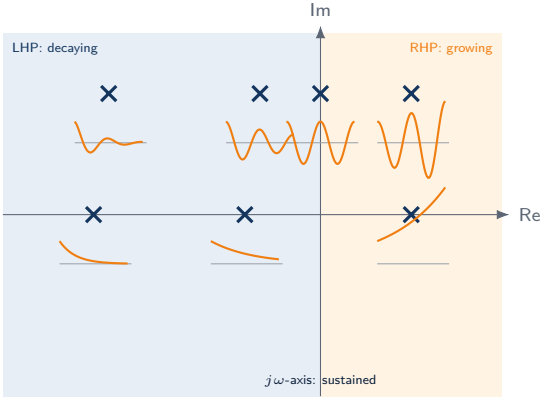



Underdamped solution ($\zeta < 1$):

$$y(t) = K\left[1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\sin(\omega_d t + \arccos\zeta)\right]$$

- Lower $\zeta$: faster rise but more overshoot and ringing.
- $\zeta \approx 0.7$: fast, $\approx 4.6\,\%$ overshoot – the classical compromise.
- $\zeta \geq 1$: no overshoot; $\zeta = 2$ is already sluggish (dominated by the slow pole $-0.27\omega_n$).
- Gray band: $\pm 2\,\%$ settling tolerance.

# Pole location versus response




Each pole $p = -\sigma \pm j\omega_d$ contributes a mode $e^{-\sigma t}\cos(\omega_d t + \phi)$:

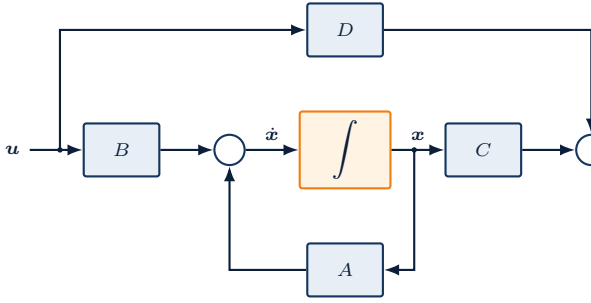
- **Real part** $-\sigma$: decay rate; time constant $1/\sigma$; 2 % settling $\approx 4/\sigma$.
- **Imaginary part** $\omega_d$: oscillation frequency.
- Farther left $\Rightarrow$ faster; but needs more actuator effort.
- **Dominant poles**: the slowest (closest to the $j\omega$-axis) dominate the response; poles $\geq 5\times$ farther left are often negligible.
- A pole–zero pair close together nearly cancels (but a slow *uncancelled* remnant may give a long "tail").

# State-space representation

$$\dot{\boldsymbol{x}} = A\boldsymbol{x} + B\boldsymbol{u}, \qquad \boldsymbol{y} = C\boldsymbol{x} + D\boldsymbol{u}$$

| Matrix | Size | Meaning |
|---|---|---|
| $A$ | $n \times n$ | system (dynamics) matrix |
| $B$ | $n \times m$ | input matrix |
| $C$ | $p \times n$ | output matrix |
| $D$ | $p \times m$ | direct feed-through |

- **State $\boldsymbol{x}$**: minimal set of variables that, with $\boldsymbol{u}(t)$, determines the future. Natural choice: energy-storage variables (inductor currents, capacitor voltages, positions, velocities).
- Handles MIMO, initial conditions and internal signals directly; basis for observers, LQR and Kalman filters (Day 3–4).
- Solution: $\boldsymbol{x}(t) = e^{At}\boldsymbol{x}(0) + \int_0^t e^{A(t-\tau)} B\boldsymbol{u}(\tau) d\tau$.




Integrators hold the state: one integrator per state variable. Reading a block diagram "integrator by integrator" gives a state-space model directly.

# Conversions between transfer function and state space

## SS $\to$ TF (unique)

$$G(s) = C\,(sI - A)^{-1}B + D = \frac{C\ \mathrm{adj}(sI - A)\,B}{\det(sI - A)} + D$$

- Poles of $G \subseteq$ eigenvalues of $A$ (equal if the realisation is **minimal**: controllable and observable).
- $D \neq 0 \Leftrightarrow G$ is biproper (no roll-off).

## TF $\to$ SS (not unique)

Any $T$ invertible: $\bar{\boldsymbol{x}} = T\boldsymbol{x}$ gives $(TAT^{-1}, TB, CT^{-1}, D)$ with the same $G(s)$.

**Controllable canonical form** for $G(s) = \dfrac{b_1 s + b_0}{s^2 + a_1 s + a_0}$:

$$A = \begin{bmatrix} 0 & 1 \\ -a_0 & -a_1 \end{bmatrix},\ B = \begin{bmatrix} 0 \\ 1 \end{bmatrix},\ C = \begin{bmatrix} b_0 & b_1 \end{bmatrix},\ D = 0$$

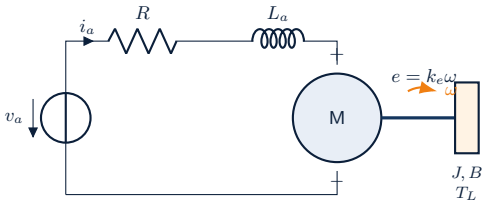
**Check:** $\det(sI - A) = s^2 + a_1 s + a_0$ ✓

**Example:** standard 2nd-order form ($K = 1$): $a_1 = 2\zeta\omega_n$, $a_0 = \omega_n^2$, $b_1 = 0$, $b_0 = \omega_n^2$:

$$A = \begin{bmatrix} 0 & 1 \\ -\omega_n^2 & -2\zeta\omega_n \end{bmatrix},\ C = \begin{bmatrix} \omega_n^2 & 0 \end{bmatrix}$$

Numerically: MATLAB/Python `tf2ss`, `ss2tf`. Use balanced or physical coordinates for implementation – canonical forms are badly conditioned for high order.

# The DC motor in both representations




$$L_a \tfrac{di_a}{dt} = v_a - Ri_a - k_e\,\omega$$
$$J \tfrac{d\omega}{dt} = k_t\, i_a - B\omega - T_L$$

SI units: $k_t$ [Nm/A] $= k_e$ [V s/rad].

**State space**, $\boldsymbol{x} = [i_a,\ \omega]^\top$, $\boldsymbol{u} = [v_a,\ T_L]^\top$, $y = \omega$:

$$A = \begin{bmatrix} -\frac{R}{L_a} & -\frac{k_e}{L_a} \\ \frac{k_t}{J} & -\frac{B}{J} \end{bmatrix},\ B = \begin{bmatrix} \frac{1}{L_a} & 0 \\ 0 & -\frac{1}{J} \end{bmatrix},\ C = \begin{bmatrix} 0 & 1 \end{bmatrix}$$

**Transfer functions** (from $C(sI - A)^{-1}B$):

$$\frac{\Omega(s)}{V_a(s)} = \frac{k_t}{(L_a s + R)(Js + B) + k_t k_e}, \qquad \frac{\Omega(s)}{T_L(s)} = \frac{-(L_a s +}{(L_a s + R)(Js +}$$

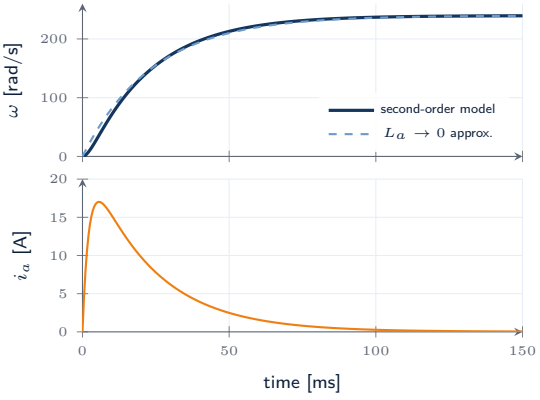
- $k_t k_e$ term = **back-EMF feedback**: the motor is an internal feedback loop.
- If $L_a \to 0$: $\Omega/V_a \approx \dfrac{1/k_e}{\tau_m s + 1}$, $\tau_m = \dfrac{RJ}{k_t k_e + RB}$ (mechanical time constant).

# DC motor: worked numerical example

Small servo motor: $R = 1.2\,\Omega$, $L_a = 2.4\,\text{mH}$, $k_t = k_e = 0.1\,\text{Nm/A}$, $J = 2 \times 10^{-4}\,\text{kg m}^2$, $B = 10^{-5}\,\text{Nm s/rad}$.

$$\det(sI - A) = s^2 + 500.05\,s + 20\,858$$

- Poles: $p_1 = -45.9\,\text{s}^{-1}$ (mechanical), $p_2 = -454\,\text{s}^{-1}$ (electrical) – both real.
- Compare $\tau_m = RJ/(k^2 + RB) = 24.0\,\text{ms}$ $(1/\tau_m = 41.7)$ and $\tau_e = L_a/R = 2\,\text{ms}$ $(1/\tau_e = 500)$: good estimates since $\tau_m \gg \tau_e$.
- DC gain $k_t/(RB + k_t k_e) = 9.99\,\text{rad/s per V} \Rightarrow$ 24 V step: 240 rad/s (2290 rpm).
- Inrush current $\approx 24/1.2 = 20\,\text{A}$, limited by $L_a$ only for $\sim \tau_e$.



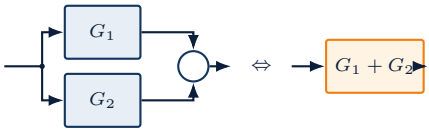

24 V voltage step, no load; computed from the state-space model.

# Block-diagram algebra

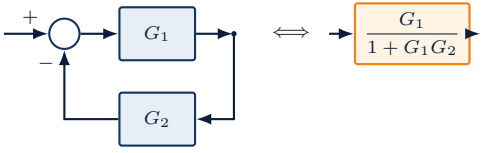
**Series**



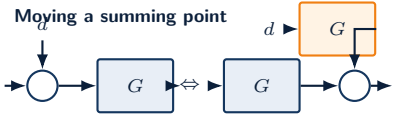

**Parallel**




**Negative feedback**




**Moving a summing point**




**Mason's rule** generalises these: $T = \frac{1}{\Delta} \sum_k P_k \Delta_k$ (forward paths $P_k$, graph determinant $\Delta$).

Positive feedback: $G_1/(1 - G_1 G_2)$. Applying rules blindly can hide **unstable pole–zero cancellations** – check internal stability.

# Session 2.2 – Summary

**Key take-aways**

- The Laplace transform makes LTI dynamics algebraic: $G(s) = Y(s)/U(s)$ with zero initial conditions.
- **Poles** decide stability and speed (modes); **zeros** shape the response; RHP zeros/delays limit performance.
- Standard forms: first order $(K, \tau)$ and second order $(K, \zeta, \omega_n)$ cover most servo sub-loops.
- State space $(A, B, C, D)$ is the internal, MIMO-ready description; $G(s) = C(sI - A)^{-1}B + D$.
- DC motor: back-EMF is an inherent feedback; electrical and mechanical poles are usually well separated.
- Block-diagram algebra: series, parallel, feedback $G_1/(1 + G_1G_2)$ – but beware hidden cancellations.

Session 3

# 2.3 System Identification and Frequency Response

# Where do models come from? Physics versus measurement

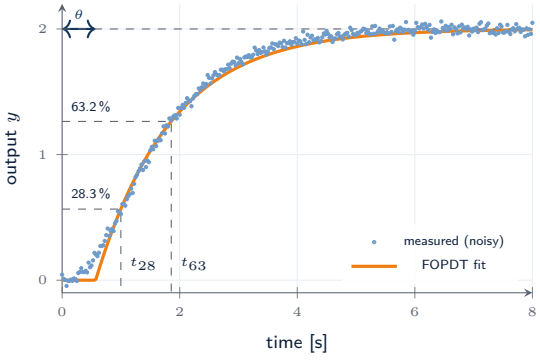
| | **White box** (first principles) | **Grey box** | **Black box** (data-driven) |
|---|---|---|---|
| Structure | from physics (Newton, Kirchhoff) | physics structure | generic (ARX, FIR, FRF) |
| Parameters | data sheets, CAD | **estimated** from data | estimated from data |
| Effort | modelling expertise | moderate | experiment design |
| Insight | high: parameters have meaning | high | low |
| Validity | wide, if assumptions hold | wide | only near tested conditions |
| Servo example | DC motor model of 2.2 | $J$, $B$, $F_C$ from run-up tests | measured FRF of a machine axis |

## Identification workflow

1. choose model structure 2. design the experiment (excitation, sample rate, amplitude) 3. collect data 4. estimate parameters 5. **validate** on fresh data 6. iterate.

Rule: the model only needs to be accurate where it matters for control – around the intended crossover frequency. Ljung's dictum: "a model is never true, only useful" [Ljung, 1999].

# Step-response identification: first order plus dead time (FOPDT)




$$G(s) = \frac{K\, e^{-\theta s}}{\tau s + 1}$$

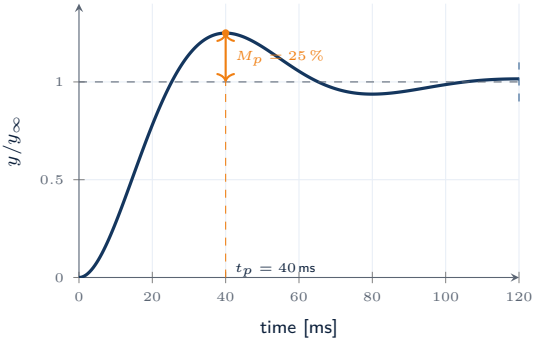
**Two-point method** (Smith): read $t_{28}$ and $t_{63}$ at 28.3 % and 63.2 % of the final change $\Delta y$:

$$K = \frac{\Delta y}{\Delta u}, \quad \tau = 1.5\,(t_{63} - t_{28}), \quad \theta = t_{63} - \tau$$

**Example** (unit step, true plant $\frac{2}{(s+1)(0.5s+1)(0.25s+1)}$): $t_{28} = 1.00\,\text{s}$, $t_{63} = 1.86\,\text{s} \Rightarrow K = 2$, $\tau = 1.29\,\text{s}$, $\theta = 0.57\,\text{s}$.

- The dead time $\theta$ absorbs the higher-order lags.
- Good for process-type and thermal loops; too coarse for resonant servo mechanics.

# Step-response identification: underdamped second order




Read the **peak overshoot** $M_p$ and the **peak time** $t_p$ (or the period $T_d$ of the ringing):

$$\zeta = \frac{-\ln M_p}{\sqrt{\pi^2 + \ln^2 M_p}}, \qquad \omega_d = \frac{\pi}{t_p} = \frac{2\pi}{T_d}, \qquad \omega_n = \frac{\omega_d}{\sqrt{1 -}}$$

**Example** (closed speed loop): $M_p = 0.25$, $t_p = 40\,\text{ms}$

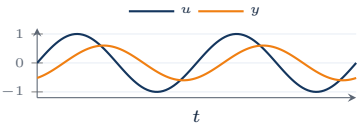
- $\ln M_p = -1.386 \Rightarrow$
  $\zeta = 1.386/\sqrt{9.870 + 1.922} = 0.404$
- $\omega_d = \pi/0.04 = 78.5\,\text{rad/s}$,
  $\omega_n = 78.5/0.915 = 85.9\,\text{rad/s}$ (13.7 Hz)

Alternative: **logarithmic decrement** $\delta = \ln(x_k/x_{k+1})$ of successive peaks, $\zeta = \delta/\sqrt{4\pi^2 + \delta^2}$ – robust for lightly damped resonances.

# Frequency response and the Bode diagram

Stable LTI system, input $u = \hat{u} \sin \omega t \Rightarrow$ in steady state

$$y = |G(j\omega)|\, \hat{u}\, \sin\Big(\omega t + \angle G(j\omega)\Big)$$



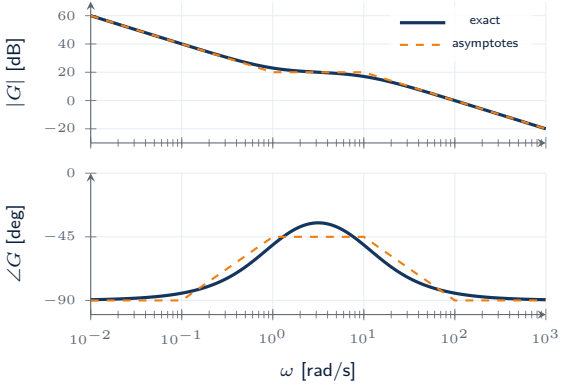

**Bode asymptote rules** (log $\omega$, dB $= 20 \log_{10} |G|$):

- each real pole: slope $-20$ dB/dec after its corner, phase $0 \to -90^\circ$ over $[0.1, 10]\times$ corner;
- each real zero: $+20$ dB/dec, phase $0 \to +90^\circ$;
- integrator $1/s$: $-20$ dB/dec everywhere, $-90^\circ$.



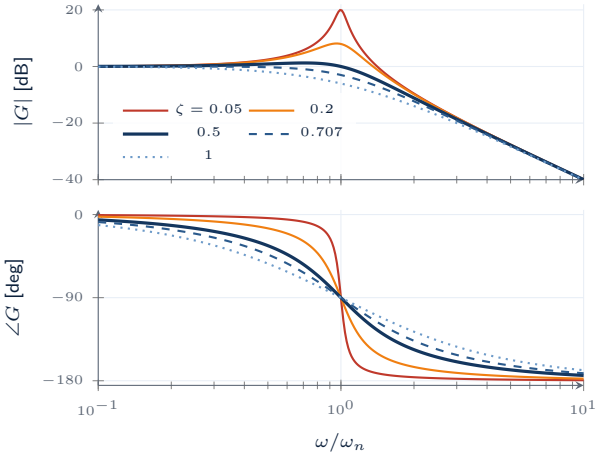
H. W. Bode (1905–1982), Bell Labs. Unknown author, Public domain, via Wikimedia Commons

$$G(s) = \frac{100(s+1)}{s(s+10)}$$




Low frequency: $G \approx 10/s$ (40 dB at 0.1 rad/s); zero at 1, pole at 10 rad/s.

# Second-order frequency response: resonance



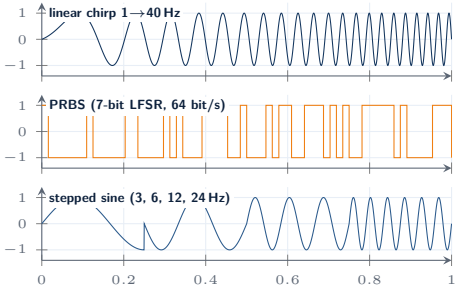

$$G(j\omega) = \frac{1}{1 - (\omega/\omega_n)^2 + j2\zeta\,\omega/\omega_n}$$

- Asymptotes: 0 dB, then $-40$ dB/dec beyond $\omega_n$; phase $0 \to -180^\circ$; at $\omega_n$: $-90^\circ$, $|G| = 1/(2\zeta)$.
- Resonance peak for $\zeta < 0.707$:

$$M_r = \frac{1}{2\zeta\sqrt{1-\zeta^2}}, \qquad \omega_r = \omega_n\sqrt{1-2\zeta^2}$$

- $\zeta = 0.05$: $M_r = 10.0$ ($+20$ dB) – typical of a **coupling or belt resonance**.
- Half-power bandwidth of the peak: $\Delta\omega \approx 2\zeta\omega_n \Rightarrow$ estimate $\zeta$ from a measured FRF.

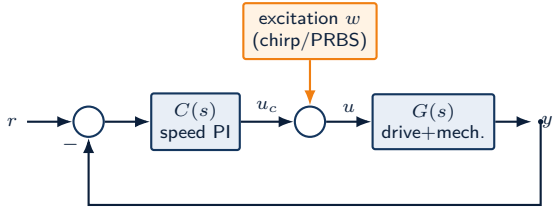
# Excitation signals for frequency-response measurement




| Signal | Pro | Con |
|---|---|---|
| Stepped sine | best SNR per line, nonlinearity visible | slow |
| Chirp / sweep PRBS | fast, smooth, low crest factor flat spectrum to $\approx 0.44 f_{\text{bit}}$, periodic $\Rightarrow$ no leakage | transient leakage; windowing binary $\Rightarrow$ excites backlash/friction |
| Multisine | designed spectrum, periodic | crest factor design needed |

- Amplitude: large enough to dominate noise and **break friction**, small enough to avoid saturation.
- Use **periodic** excitation and an integer number of periods to avoid leakage.
- Average several periods/blocks to reduce noise variance.

# FRF measurement in servo drives




| Ratio of spectra | Gives | Remark |
|---|---|---|
| $\hat{G} = \hat{S}_{wy}/\hat{S}_{wu}$ | plant $G$ | robust in closed loop |
| $\hat{S}_{wy}/\hat{S}_{ww}$ | process sensitivity $GS$ | |
| $-\hat{S}_{wu_c}/\hat{S}_{ww}$ | $T$ (input side) | |
| $\hat{S}_{uy}/\hat{S}_{uu}$ $(H_1)$ | $G$ only if open loop | **biased** in closed loop |

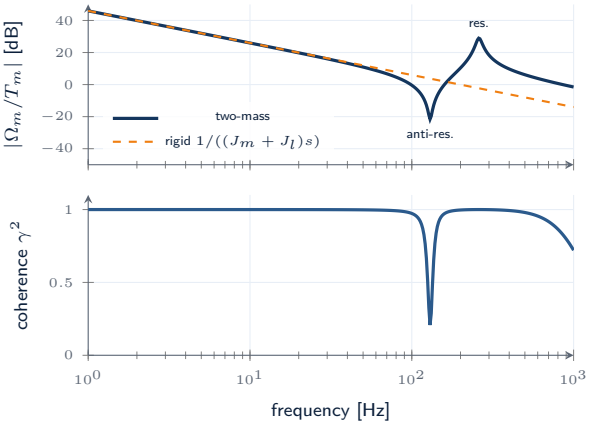


Loxyger, CC BY 4.0, via Wikimedia Commons

HP 3562A dynamic signal analyser (1985): the classic tool. Modern servo drives have the analyser **built in**: the drive injects $w$ and records $u, y$ at the loop sample rate.

Measure in **closed loop** (axis must hold position, gravity, safety) and compute $G$ from the cross-spectra with $w$.

# Measured FRF of a two-mass axis and coherence




Motor inertia $J_m = 2 \times 10^{-4}$, load $J_l = 6 \times 10^{-4}\,\mathrm{kg\,m^2}$, shaft stiffness $k = 400\,\mathrm{Nm/rad}$:

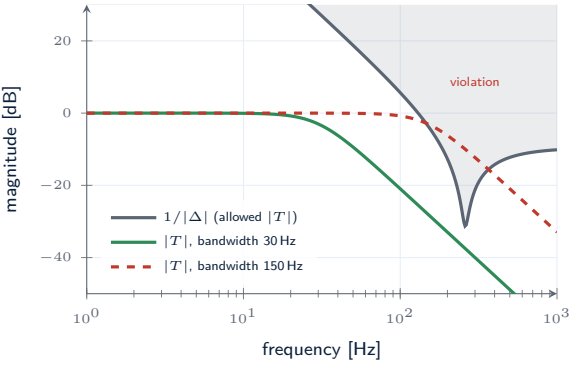
- anti-resonance $f_{ar} = \frac{1}{2\pi}\sqrt{k/J_l} = 130\,\mathrm{Hz}$
- resonance $f_r = \frac{1}{2\pi}\sqrt{k\,\frac{J_m+J_l}{J_m J_l}} = 260\,\mathrm{Hz}$

  $(f_r/f_{ar} = \sqrt{1 + J_l/J_m} = 2)$

$$\gamma^2(\omega) = \frac{|\hat{S}_{uy}|^2}{\hat{S}_{uu}\hat{S}_{yy}} \in [0, 1]$$

- $\gamma^2 \approx 1$: output linearly explained by input. Low $\gamma^2$: noise, nonlinearity, leakage.
- Coherence drops at the **anti-resonance** (tiny output) and at high frequency ⇒ trust $\hat{G}$ only where $\gamma^2 \gtrsim 0.8$.

# Model uncertainty




Every model is wrong somewhere. Describe the error as a **multiplicative perturbation**:

$$G_{\text{true}}(s) = G_0(s)\big(1 + \Delta(s)\big)$$

Here $G_0$ = rigid-body model, $G_{\text{true}}$ = two-mass system of the previous slide.

## Robust stability (small-gain)

If the nominal loop is stable, it stays stable for all $\Delta$ with $|\Delta(j\omega)| \le |W(j\omega)|$ iff

$$|T(j\omega)|\,|W(j\omega)| < 1 \quad \forall\omega$$

- Relative error is small at low frequency and $> 100\,\%$ near resonances.
- $\Rightarrow$ The bandwidth must stay **below the first mechanical resonance** unless the resonance is modelled (notch filters, Day 3).

# Session 2.3 – Summary

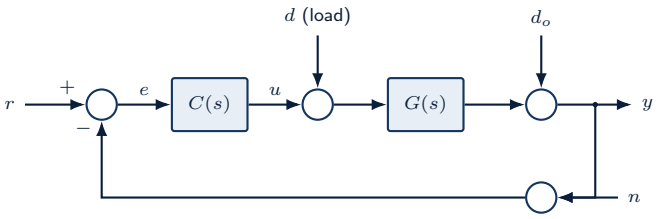
**Key take-aways**

- Combine physics (structure) with measurement (parameters): grey-box models are the servo engineer's default.
- Step tests give FOPDT ($K, \tau, \theta$) or 2nd-order ($\zeta, \omega_n$ from $M_p$, $t_p$) models quickly.
- The Bode diagram is built from asymptotes: $\pm 20\,\text{dB/dec}$ per real pole/zero, $-40\,\text{dB/dec}$ and a peak $1/(2\zeta)$ per resonant pair.
- FRF measurement: periodic excitation (chirp, PRBS, multisine), averaging, cross-spectra with the injected signal in closed loop.
- Check **coherence**; quantify model error $\Delta(j\omega)$ – it limits the achievable bandwidth.

Session 4

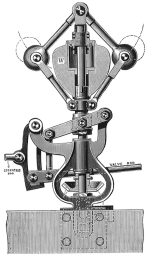
# 2.4 PID and Basic Control Structures

# The standard feedback loop




$$Y = \underbrace{\frac{L}{1+L}}_{T} R + \underbrace{\frac{G}{1+L}}_{GS} D + \underbrace{\frac{1}{1+L}}_{S} D_o - T\,N$$

- Large $|L| \Rightarrow S \to 0$: disturbances rejected, $y$ follows $r$ – **but** $T \to 1$: noise passes to $y$.
- Four "gang of four" transfer functions $S$, $T$, $GS$, $CS$ must *all* be stable [Åström & Murray, 2008].

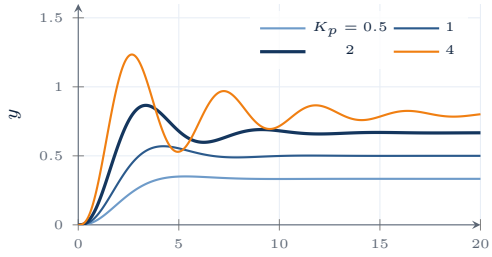


Heat Engines (1913), scan by Andy Dingley, Public domain, via Wikimedia Commons

Watt's centrifugal governor: a mechanical **proportional** speed controller; Maxwell's 1868 analysis "On Governors" started control theory.

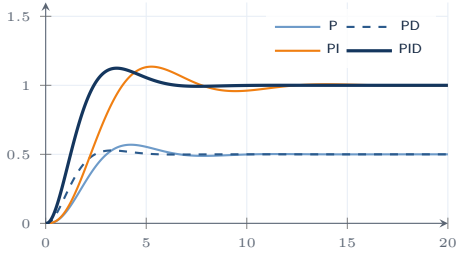
# Effects of P, I and D actions

P only, $G = 1/(s+1)^3$




Offset $e_\infty = 1/(1+K_p)$; $K_p = 8 \Rightarrow$ sustained oscillation.

same plant, four controllers




P: $K_p = 1$; PI: $K_p = 1, T_i = 2$; PD: $K_p = 1, T_d = 0.8$; PID: $K_p = 2, T_i = 2, T_d = 0.6$ ($N = 10$).

| Increase ... | rise time | overshoot | steady-state error | stability margin |
|---|---|---|---|---|
| $K_p$ | faster | larger | smaller (not zero) | lower |
| $1/T_i$ (I action) | faster | larger | **eliminated** (step) | lower (phase lag) |
| $T_d$ (D action) | little change | smaller | no effect | higher (phase lead), but noise ↑ |

# PID forms: parallel, ideal (ISA) and series

| Form | Transfer function | Typical use |
|---|---|---|
| Parallel | $C = K_p + \dfrac{K_i}{s} + K_d s$ | textbooks, drives |
| Ideal (ISA) | $C = K_p\left(1 + \dfrac{1}{T_i s} + T_d s\right)$ | process controllers |
| Series (interacting) | $C = K_p'\left(1 + \dfrac{1}{T_i' s}\right)\left(1 + T_d' s\right)$ | legacy, PLCs |

**Conversions:** $K_i = K_p/T_i$, $K_d = K_p T_d$; series $\rightarrow$ ideal:

$$K_p = K_p' \frac{T_i' + T_d'}{T_i'}, \quad T_i = T_i' + T_d', \quad T_d = \frac{T_i' T_d'}{T_i' + T_d'}$$

- The series form always has real zeros (possible only if $T_i \geq 4T_d$ in ideal form).
- Changing $K_p$ in the ideal form scales all three actions; in the parallel form only P.
- **Pitfall**: copying gains between devices without checking the form and units (s vs. min, % vs. engineering units, rad vs. rev).



Bitjungle, CC BY-SA 4.0, via Wikimedia Commons

Industrial PID controllers: tuning parameters are usually entered as $K_p$ (or proportional band $100/K_p$ %), $T_i$, $T_d$ in the ideal form.

# Classical tuning: Ziegler–Nichols (1942)

**Ultimate-sensitivity method:** P-only, raise $K_p$ until sustained oscillation: ultimate gain $K_u$, period $T_u$.

| | $K_p$ | $T_i$ | $T_d$ |
|---|---|---|---|
| P | $0.5K_u$ | – | – |
| PI | $0.45K_u$ | $T_u/1.2$ | – |
| PID | $0.6K_u$ | $T_u/2$ | $T_u/8$ |

**Step-response method:** FOPDT $(K, \tau, \theta)$, $a = K\theta/\tau$.

| | $K_p$ | $T_i$ | $T_d$ |
|---|---|---|---|
| P | $1/a$ | – | – |
| PI | $0.9/a$ | $3\theta$ | – |
| PID | $1.2/a$ | $2\theta$ | $\theta/2$ |

## Worked example

Plant $G = 1/(s+1)^3$. Phase $-180^\circ$ when $3\arctan\omega = 180^\circ \Rightarrow \omega_u = \sqrt{3}$ rad/s; $|G(j\omega_u)| = 1/8 \Rightarrow K_u = 8$, $T_u = 2\pi/\sqrt{3} = 3.63$ s.
ZN PID: $K_p = 4.8$, $T_i = 1.81$ s, $T_d = 0.45$ s.

## Critique

- Designed for **quarter-amplitude decay** ($\zeta \approx 0.2$): aggressive, $M_s$ often $> 3$.
- Pushing a real machine to the stability limit can be unsafe.
- Use ZN only as a starting point; modern rules (SIMC, AMIGO, lambda) are more robust [Åström & Hägglund, 2006].

# Model-based tuning: lambda, SIMC, Cohen–Coon

For FOPDT $G = Ke^{-\theta s}/(\tau s + 1)$, PI controller $K_p(1 + 1/(T_i s))$:

| Method | $K_p$ | $T_i$ |
|---|---|---|
| Lambda (IMC) | $\dfrac{\tau}{K(\lambda + \theta)}$ | $\tau$ |
| SIMC [Skogestad 2003] | $\dfrac{\tau}{K(\tau_c + \theta)}$ | $\min\{\tau,\ 4(\tau_c + \theta)\}$ |
| Cohen–Coon | $\dfrac{\tau}{K\theta}\left(0.9 + \dfrac{\theta}{12\tau}\right)$ | $\theta\dfrac{30 + 3\theta/\tau}{9 + 20\theta/\tau}$ |

- One intuitive knob: the desired closed-loop time constant $\lambda$ or $\tau_c$.
- SIMC default $\tau_c = \theta$: good trade-off, $M_s \approx 1.6$–$1.7$.
- Lambda tuning: $T_i = \tau$ cancels the plant pole $\Rightarrow$ slow load-disturbance recovery if $\tau \gg \theta$.

## Example: FOPDT of session 2.3

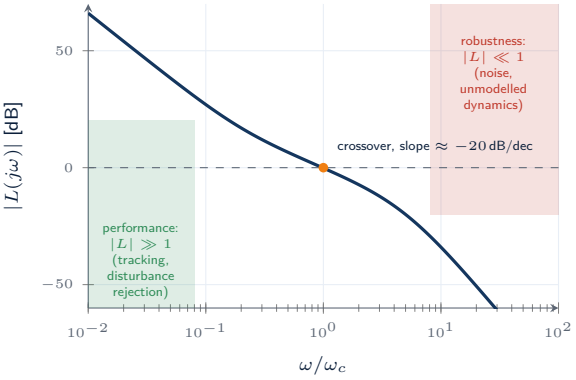
$K = 2$, $\tau = 1.29\,\text{s}$, $\theta = 0.57\,\text{s}$ ($\theta/\tau = 0.44$)

| Rule | $K_p$ | $T_i$ [s] |
|---|---|---|
| ZN step (PI) | 1.02 | 1.71 |
| Cohen–Coon (PI) | 1.06 | 1.00 |
| SIMC, $\tau_c = \theta$ | 0.56 | 1.29 |
| Lambda, $\lambda = \tau$ | 0.35 | 1.29 |

Spread of almost 3:1 in $K_p$ – the rules encode **different robustness targets**, not "the" answer.

Servo drives typically use model-based rules from inertia and bandwidth instead (see cascade slide).

# Loop shaping: designing $|L(j\omega)|$ directly



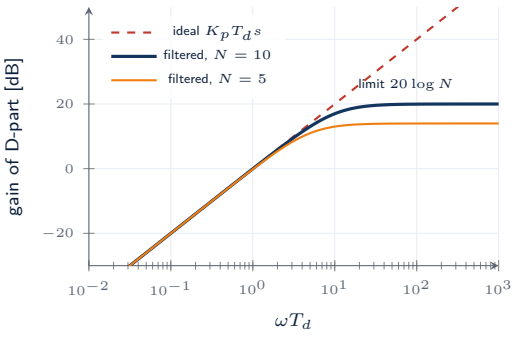

Since $S = 1/(1+L)$ and $T = L/(1+L)$:

- $|L| \gg 1 \Rightarrow |S| \approx 1/|L|$ small: good tracking and load rejection.
- $|L| \ll 1 \Rightarrow |T| \approx |L|$ small: noise and model errors suppressed.
- Around $\omega_c$: slope about $-20$ dB/dec over about a decade $\Rightarrow$ phase near $-90^\circ \ldots -130^\circ \Rightarrow$ healthy PM (Bode gain–phase relation).

## Loop-shaping with PID

I action raises low-frequency gain (slope $-40$ dB/dec below $1/T_i$); D action adds phase lead around $\omega_c$; the filter $N$ and extra low-pass filters give high-frequency roll-off. Practical rule: $1/T_i \approx \omega_c/4 \ldots \omega_c/10$.

# Derivative action: filtering and placement

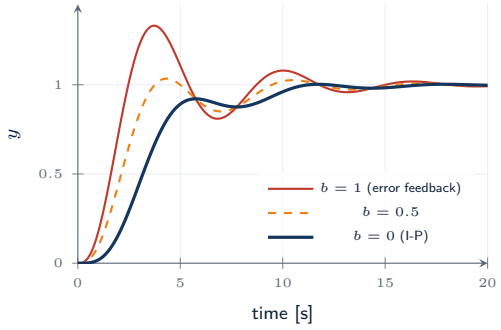



An ideal differentiator has unbounded high-frequency gain: encoder quantisation and noise are amplified.

$$C(s) = K_p\Big(1 + \frac{1}{T_i s} + \frac{T_d s}{1 + T_d s/N}\Big), \quad N \approx 5 \ldots 20$$

- High-frequency D-gain limited to $K_p N$; filter pole at $N/T_d$.
- Alternatively filter the measurement: $C(s) \cdot \frac{1}{(T_f s+1)^2}$ (2nd-order roll-off).
- **Derivative on measurement**, $u_D = -K_p T_d\, \dot{y}_f$: avoids the "derivative kick" at set-point steps.
- Servo practice: D-action on position = velocity feedback from the encoder; in cascade form this is the inner velocity loop.
- Example: 20-bit encoder, $T_s = 125\,\mu$s: one count $\Rightarrow$ velocity step of $2\pi/(2^{20} T_s) = 0.048$ rad/s per sample.

# Set-point weighting: two degrees of freedom



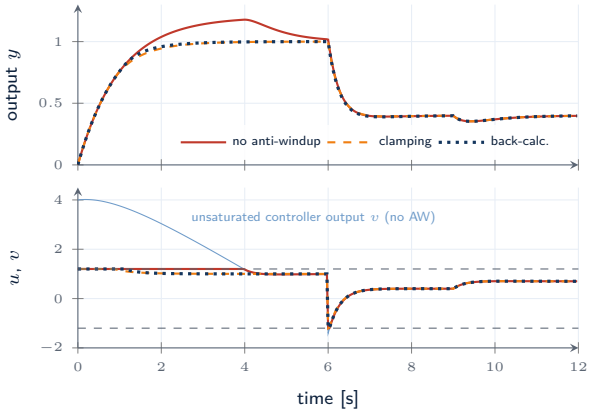

PI, $K_p = 2$, $T_i = 2.5\,\text{s}$, $G = 1/(s+1)^3$. Overshoot: 33 %, 3.4 %, 0.3 %.

Weight the set-point differently in each term:

$$u = K_p\Big[(b\,r - y) + \frac{1}{T_i}\int (r - y)\,dt + T_d\,\frac{d}{dt}(c\,r - y_f)\Big]$$

- The I-term must use the true error $(r - y)$ so the steady state is exact.
- Typical: $c = 0$ (no derivative kick), $0 \le b \le 1$.
- Feedback properties ($L$, $S$, $T$, margins, **load-disturbance response**) are *unchanged*: $b, c$ only shape the set-point response.
- Tune $K_p, T_i, T_d$ for disturbances and robustness first, then $b$ for the set-point.
- Same idea in servos: reference pre-filter or trajectory generator + feedforward.

# Actuator saturation and integrator windup

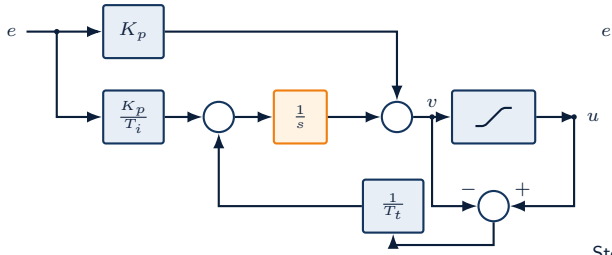



Plant $1/(s+1)$, PI $K_p = 4$, $T_i = 0.8\,\text{s}$, limit $|u| \le 1.2$. Set-point $1 \to 0.4$ at 6 s, load step at 9 s.

- While $u$ is saturated the loop is **open**: the integrator keeps integrating the error.
- The "wound-up" integrator must be unwound by an error of opposite sign $\Rightarrow$ 18 % overshoot, long settling, sluggish reaction to the set-point change.
- In servos: current limit (torque), voltage limit (DC-link), speed limit in the cascade.
- Every saturating integrator needs anti-windup – including the speed loop when the current reference is clamped.

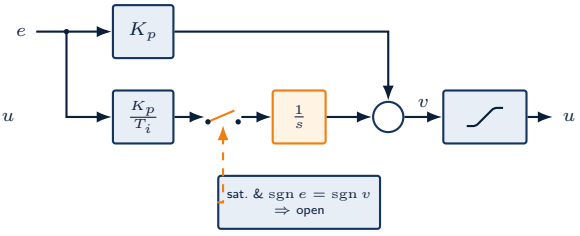
# Anti-windup schemes

**Back-calculation (tracking)**




Integrator input: $\frac{K_p}{T_i}e + \frac{1}{T_t}(u - v)$. When saturated, the integrator is driven so that $v$ tracks $u$. Choice: $T_t \approx \sqrt{T_i T_d}$, or $T_t \approx T_i/2$ for PI [Åström & Hägglund, 2006]. With a model of the limit, also works for rate limits.

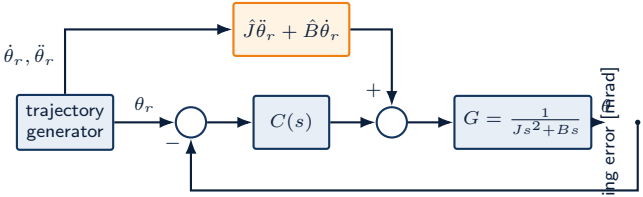
**Conditional integration (clamping)**




Stop integrating when the output is saturated *and* the error would drive it further into saturation. Simple, no extra parameter; standard in drive firmware.

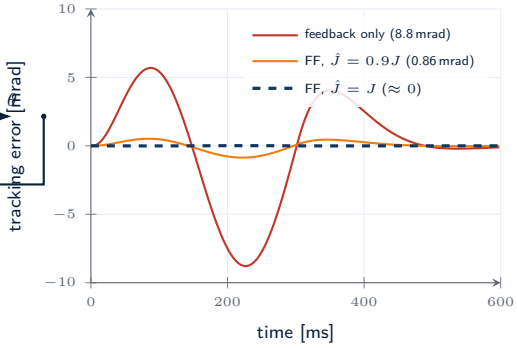
Other: integrator **clamping to limits**, incremental (velocity) form $\Delta u_k$ with output limiting, observer-based anti-windup. In cascades, pass the inner loop's saturation back to the outer loop.

# Feedforward: let feedback correct only what is unknown



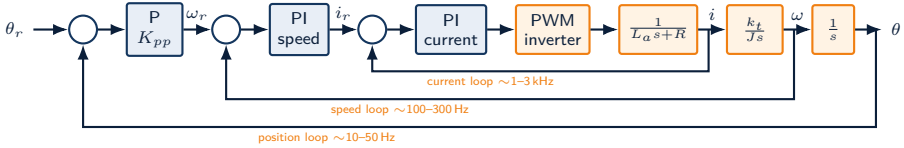

Ideal feedforward $F = G^{-1}$: then $e \equiv 0$ for a perfect model, independent of $C$. Physical meaning: torque = inertia $\times$ acceleration + friction.

- Needs a **smooth reference** (bounded jerk) $\Rightarrow$ S-curve / polynomial trajectories.
- FF does not affect stability (outside the loop); feedback handles model errors and disturbances.



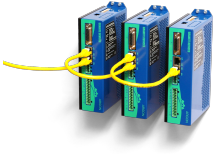

1 rad move in 300 ms, $J = 10^{-3}\,\mathrm{kg\,m^2}$, PID with $\omega_c \approx 2\pi \cdot 25\,\mathrm{rad/s}$. Even a 10 % inertia error gives a $10\times$ improvement.

# Cascade control: the servo drive structure




| Loop | Simple design rule | Example (DC motor of session 2.2) |
|---|---|---|
| Current PI | $K_p = L_a\omega_{cc}$, $T_i = L_a/R$ | 1 kHz: $K_p = 15.1\,\mathrm{V/A}$, $T_i = 2\,\mathrm{ms}$ |
| Speed PI | $K_p = J\omega_{sc}/k_t$, $T_i \approx 4/\omega_{sc}$ | 100 Hz: $K_p = 1.26\,\mathrm{A\,s/rad}$, $T_i = 6.4\,\mathrm{ms}$ |
| Position P | $K_{pp} \approx \omega_{sc}/4 \ldots /10$ | $K_{pp} = 126\,\mathrm{s}^{-1}$ (20 Hz) |

- Inner loops linearise and speed up the plant seen by the outer loop; tune **inside-out**.
- Bandwidth separation $\geq$ 4–10×; limits ($i_{\max}$, $\omega_{\max}$) are natural set-point clamps. Details on Day 4.



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# Session 2.4 – Summary

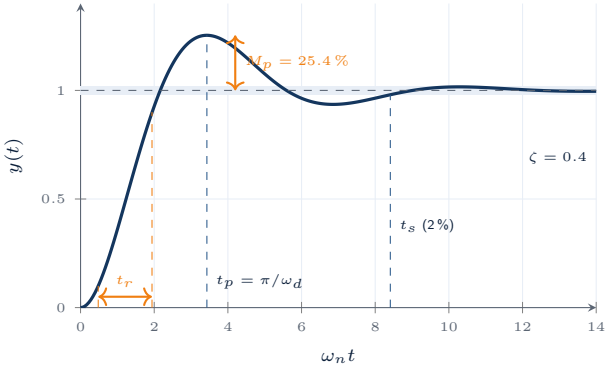
**Key take-aways**

- P sets speed, I removes steady-state error (adds lag), D adds phase lead (and noise) – always filter D.
- Know the controller **form** (parallel / ideal / series) and units before entering gains.
- Ziegler–Nichols is aggressive; model-based rules (SIMC, lambda) expose one robustness knob.
- Set-point weighting and feedforward improve tracking *without* changing feedback robustness.
- Every saturating integrator needs **anti-windup** (clamping or back-calculation).
- Servo drives: cascaded current / speed / position loops, tuned inside-out with bandwidth separation.

Session 5

# 2.5 Stability and Performance Analysis

# Time-domain specifications

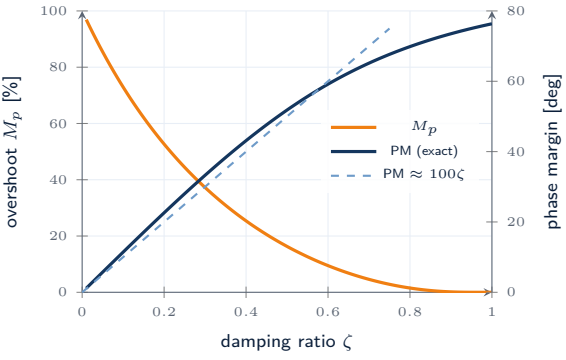



| Measure | Definition |
|---|---|
| Rise time $t_r$ | $10\,\% \to 90\,\%$ of final value |
| Peak time $t_p$ | time of first maximum |
| Overshoot $M_p$ | $(y_{\max} - y_\infty)/y_\infty$ |
| Settling time $t_s$ | last entry into $\pm 2\,\%$ band |
| Steady-state error | $e_\infty = \lim_{t\to\infty} e(t)$ |

For $\zeta = 0.4$: $\omega_n t_r = 1.46$, $\omega_n t_p = 3.43$, $\omega_n t_s = 8.4$ (estimate $4/\zeta = 10$).

> **Transient** = response to the initial shock (poles). **Steady state** = what remains (system type, DC gain). In servos also: following error during constant velocity, in-position window and settling time to it.

# Specifications versus damping ratio




For the closed-loop second-order system (exact formulas):

$$M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}}, \quad t_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}, \quad t_s \approx \frac{4}{\zeta\omega_n}$$

Rules of thumb: $t_r \approx 1.8/\omega_n$ (for $\zeta \approx 0.5$), $\omega_b \approx \omega_n$ for $\zeta = 0.707$.

Link to the open loop $L(s) = \omega_n^2/(s(s+2\zeta\omega_n))$:

$$\mathrm{PM} = \arctan \frac{2\zeta}{\sqrt{\sqrt{1+4\zeta^4} - 2\zeta^2}} \approx 100\,\zeta \text{ [deg]}$$

| $\zeta$ | 0.3 | 0.4 | 0.5 | 0.6 | 0.7 |
|---|---|---|---|---|---|
| $M_p$ [%] | 37 | 25 | 16 | 9.5 | 4.6 |
| PM [deg] | 33 | 43 | 52 | 59 | 65 |

# Steady-state error, system type and the final value theorem

Unity feedback, $E(s) = S(s)R(s) = \dfrac{R(s)}{1 + L(s)}$. If $sE(s)$ has all poles in the LHP:

$$e_\infty = \lim_{t\to\infty} e(t) = \lim_{s\to 0} s\,E(s)$$

**System type** = number of integrators in $L(s)$:
$L(s) = \dfrac{K\,\tilde{L}(s)}{s^N}$, $\tilde{L}(0) = 1$.

Error constants: $K_p = \lim_{s\to 0} L$, $K_v = \lim_{s\to 0} sL$, $K_a = \lim_{s\to 0} s^2 L$.

**Pitfall**: FVT applied to an unstable loop gives a finite but meaningless number.

| Type $N$ | step $r = 1$ | ramp $r = t$ | parabola $r = t^2/2$ |
|---|---|---|---|
| 0 | $\dfrac{1}{1+K_p}$ | $\infty$ | $\infty$ |
| 1 | 0 | $\dfrac{1}{K_v}$ | $\infty$ |
| 2 | 0 | 0 | $\dfrac{1}{K_a}$ |

**Servo example**: position loop with P gain $K_{pp} = 126\,\mathrm{s}^{-1}$ around an ideal velocity loop $\Rightarrow L = K_{pp}/s$, type 1. At $v = 0.2\,\mathrm{m/s}$ on a 10 mm-pitch ball screw ($\dot\theta = 126\,\mathrm{rad/s}$): following error $e_\infty = \dot\theta/K_v = 126/126 = 1\,\mathrm{rad} = 1.6\,\mathrm{mm}$.

- Velocity feedforward removes it (makes the reference path "type 2").
- Integral action in the position loop adds a type but also phase lag.

# Routh–Hurwitz criterion

Stability without computing roots: for $a_n s^n + a_{n-1} s^{n-1} + \cdots + a_0$

- Necessary: all coefficients present and of the same sign.
- Build the Routh array; number of **sign changes** in the first column = number of RHP roots.

Each new entry: $b_1 = \dfrac{a_{n-1}a_{n-2} - a_n a_{n-3}}{a_{n-1}}$, etc.

**Example:** $L(s) = \dfrac{K}{s(s+2)(s+5)}$, unity feedback:

$$1 + L = 0 \;\Leftrightarrow\; s^3 + 7s^2 + 10s + K = 0$$

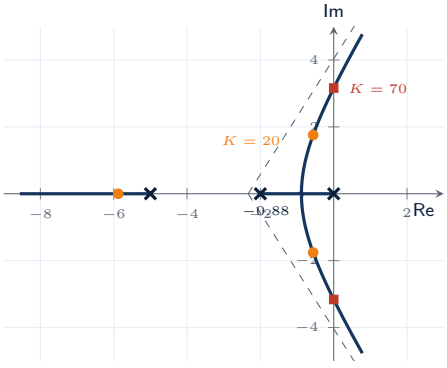
| | | |
|---|---|---|
| $s^3$ | $1$ | $10$ |
| $s^2$ | $7$ | $K$ |
| $s^1$ | $\dfrac{70 - K}{7}$ | $0$ |
| $s^0$ | $K$ | |

**Result**

Stable $\Leftrightarrow 0 < K < 70$.
At $K = 70$: auxiliary polynomial $7s^2 + 70 = 0 \Rightarrow s = \pm j\sqrt{10} = \pm j3.16$ rad/s: the loop oscillates at the ultimate frequency. For $K = 20$: gain margin $70/20 = 3.5$ (10.9 dB).

Limitation: yes/no answer, no direct information on damping or robustness; does not handle delays.

# Root locus: how closed-loop poles move with gain

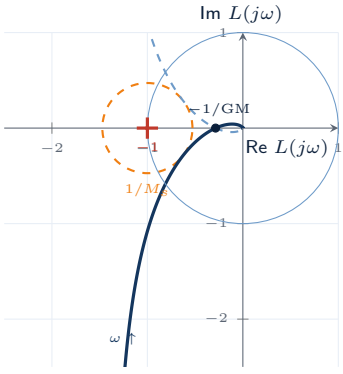



Locus of the roots of $1 + K\,\tilde{L}(s) = 0$ for $K : 0 \to \infty$; here $\tilde{L} = \dfrac{1}{s(s+2)(s+5)}$.

- Starts at open-loop **poles** ($K = 0$), ends at open-loop zeros or at infinity.
- On the real axis left of an odd number of poles+zeros.
- Asymptotes: centroid $\sigma_a = \frac{0-2-5}{3} = -2.33$, angles $\pm 60°, 180°$.
- Break-away: $\frac{d}{ds}[s(s+2)(s+5)] = 0 \Rightarrow s = -0.88$ ($K = 4.06$).
- $j\omega$-crossing at $\pm j3.16$, $K = 70$ – consistent with Routh.
- $K = 20$: poles $-0.56 \pm j1.76$ ($\zeta = 0.30$) and $-5.88$.

**Use**: see at a glance how gain trades speed for damping, and why more gain eventually destabilises a $\geq$ 3rd-order loop.

# Nyquist criterion



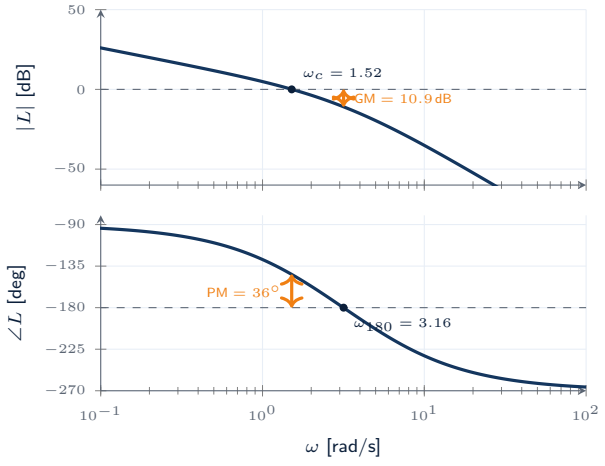

Plot $L(j\omega)$ for $\omega : -\infty \to \infty$ (indent around poles on the $j\omega$-axis).

$$Z = N + P$$

$Z$: closed-loop RHP poles, $P$: open-loop RHP poles, $N$: clockwise encirclements of $-1$.

- Stable open loop ($P = 0$): closed loop stable $\Leftrightarrow$ no encirclement of $-1$.
- Here $L = 20/(s(s+2)(s+5))$: crosses the real axis at $-20/70 = -0.286 \Rightarrow \mathrm{GM} = 3.5$.
- Unit circle crossing gives the phase margin.
- **Distance to $-1$** is the true robustness measure: $\min_\omega |1 + L| = 1/M_s$.
- Works with dead time $e^{-sT_d}$ and with measured FRF data – no model needed.

# Gain and phase margins on the Bode diagram

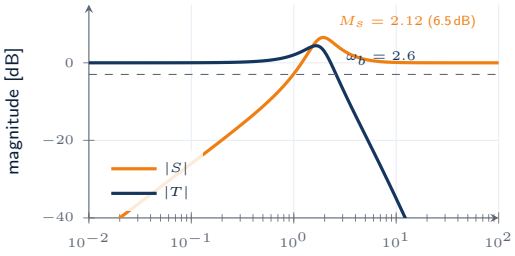


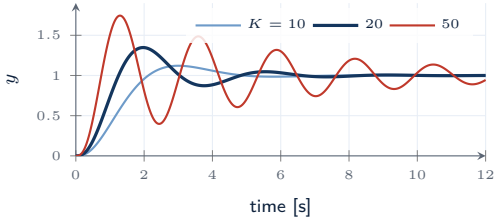

## Definitions

Crossover $\omega_c$: $|L(j\omega_c)| = 1$; PM $= 180^\circ + \angle L(j\omega_c)$
Phase crossover $\omega_{180}$: $\angle L = -180^\circ$; GM $= 1/|L(j\omega_{180})|$

- GM: how much the gain may increase (motor $k_t$, inertia drop) before instability.
- PM: how much extra **phase lag** is tolerable. A pure delay $T_d$ costs $\omega_c T_d$ rad: delay margin $= \mathrm{PM}/\omega_c$.
- Example $K = 20$: PM $= 36^\circ \Rightarrow$ delay margin $= 0.628/1.515 = 0.41\,\mathrm{s}$.
- Typical targets: PM $45$–$60^\circ$, GM $\geq$ 6–10 dB.
- Margins can mislead for loops with several crossings – check $M_s$.

# Sensitivity functions, $M_s$ and bandwidth




- $S = \frac{1}{1+L}$: disturbance/error response; $T = \frac{L}{1+L}$: tracking and noise response; $S + T = 1$.
- **Bandwidth** $\omega_b$: where $|T|$ falls to $-3\,\text{dB}$; typically $\omega_c < \omega_b < 2\omega_c$.
- $M_s = \max_\omega |S(j\omega)|$; $|S| > 1$ means feedback **amplifies** disturbances there.

$$\text{GM} \geq \frac{M_s}{M_s - 1}, \qquad \text{PM} \geq 2 \arcsin \frac{1}{2M_s}$$

- $M_s = 2$: GM $\geq 6\,\text{dB}$, PM $\geq 29^\circ$. $M_s = 1.4$: GM $\geq 10.9\,\text{dB}$, PM $\geq 41^\circ$.
- Recommended $M_s = 1.2$–$2.0$ [Åström & Hägglund, 2006].
- **Waterbed effect** (Bode integral): $\int_0^\infty \ln |S(j\omega)|\, d\omega = 0$ for stable $L$ with relative degree $\geq 2$ – pushing $|S|$ down somewhere raises it elsewhere.

# Session 2.5 – Summary and design targets

## Key take-aways

- Specs: $t_r$, $t_p$, $M_p$, $t_s$, $e_\infty$; for 2nd order all follow from $\zeta$ and $\omega_n$.
- System type decides steady-state error; FVT only for stable loops.
- Routh: yes/no stability and gain limits; root locus: pole movement with gain.
- Nyquist: encirclements of $-1$; works with delays and measured FRFs.
- Margins PM, GM and above all $M_s$ quantify robustness; $\omega_c$, $\omega_b$ quantify speed.

**Typical design targets (servo / process)**

| Quantity | Target |
|---|---|
| Phase margin PM | 45–60° |
| Gain margin GM | 6–12 dB |
| Max. sensitivity $M_s$ | 1.3–2.0 |
| Max. compl. sens. $M_t$ | $\leq 1.3 \ldots 1.5$ |
| Overshoot (position step) | 0–5 % (often 0) |
| $\omega_c$ vs. first resonance | $\omega_c < \omega_r/3$ (w/o notch) |
| $\omega_c$ vs. sample rate | $\omega_c < \omega_s/20$ roughly |

Our example $K = 20$: PM $36^\circ$, $M_s = 2.12 \Rightarrow$ too aggressive; $K \approx 10$ would meet the targets.

Session 6

# Wrap-up

# Day summary

## Modelling chain

nonlinear physics $\xrightarrow{\text{Jacobian}}$ $(A, B, C, D)$ $\xrightarrow{C(sI-A)^{-1}B+D}$ $G(s)$ $\xrightarrow{s=j\omega}$ Bode / Nyquist

... and in the lab: step test / FRF $\rightarrow$ FOPDT, 2nd order, non-parametric $\hat{G}(j\omega)$ + uncertainty $\Delta(j\omega)$.

## Control chain

PID (form, filter, set-point weights) + anti-windup + feedforward, cascaded current $\rightarrow$ speed $\rightarrow$ position.

## Ten numbers to remember

| | |
|---|---|
| 63 % | first-order step at $t = \tau$ |
| $\zeta = 0.7$ | $\approx 5$ % overshoot, PM $\approx 65^\circ$ |
| $-20$ dB/dec | per pole; $-40$ dB/dec per pair |
| $1/(2\zeta)$ | resonance peak at $\omega_n$ |
| $\gamma^2 > 0.8$ | trustworthy FRF |
| $N = 5$–$20$ | derivative filter |
| 4–10$\times$ | cascade bandwidth separation |
| 45–60$^\circ$ | phase margin |
| 6–12 dB | gain margin |
| $M_s \leq 2$ | robustness |

# Preview of Day 3

## Day 3 – Control: filters, modern, adaptive and discrete control + servo architecture

- Filters in the loop: low-pass, high-pass and notch filters, their phase lag and placement; lead/lag compensators.
- Modern control: state feedback, pole placement, controllability, observability, observers.
- Adaptive and robust control: parameter variation, MRAC structure, Lyapunov reasoning, goals of robust control.
- Discrete-time control: sampling, zero-order hold, discretisation, computation delay, stability in the $z$-plane.
- Start of servo systems: architecture, power electronics – inverter and DC link.

**Bridge from today**

- State space $(A, B, C, D) \rightarrow$ state feedback and observers.
- Phase margin and $M_s \rightarrow$ the phase cost of every filter.
- Delay margin $\rightarrow$ sampling and computation delay.
- Uncertainty $\Delta(j\omega) \rightarrow$ robust and adaptive control.



Pieceofmetalwork, CC BY-SA 4.0, via Wikimedia Commons

Strain-wave gear: compliant transmissions motivate notch filters and observers.

# Figure credits

All other drawings and plots: own work (TikZ / pgfplots; numerical data from `day2/data/gen_data.py`).

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